

DCANFL: Decentralized Federated Learning for In-Vehicle CAN Intrusion Detection

Yunjian Jia, Zirui Liu, Wanli Wen, Tian Tian, and Liang Liang

Abstract—The controller area network (CAN) bus, a core in-vehicle communication protocol, is highly vulnerable to cyber attacks due to its lack of built-in security mechanisms. While many intrusion detection systems (IDS) have been proposed, most rely on centralized training that requires sensitive data to be uploaded to a server, raising privacy and deployment concerns. Federated learning (FL) mitigates these issues by keeping data local, yet conventional FL still depends on a central coordinator, making it unsuitable for vehicular networks with intermittent connectivity and single points of failure. To address these challenges, we propose DCANFL, a decentralized FL framework for in-vehicle CAN intrusion detection. DCANFL uses gossip-based, dissimilarity-driven pairing for efficient model exchange and a continual learning training pipeline to improve performance and maintain high accuracy across diverse attack types. Experiments on two in-vehicle CAN datasets show that DCANFL, without central server coordination, achieves better detection performance than centralized FedAvg and other decentralized FL baselines.

Index Terms—Vehicle CAN bus security, Decentralized federated learning, Intrusion detection, Unsupervised learning.

I. INTRODUCTION

The rapid development of intelligent vehicles has established the Controller Area Network (CAN) as the core protocol for powertrain, steering, and infotainment systems. However, lacking built-in authentication and encryption, CAN is highly vulnerable to injection, replay, and masquerade attacks, which threaten vehicle stability and passenger safety [1]. To address these risks, intrusion detection systems (IDS) have become a major research focus [2], [3], [4]. For example, CANet uses multi-scale signal decomposition and autoencoders for unsupervised anomaly detection without labeled data, while X-CANIDS employs signal deserialization to improve detection accuracy, interpretability, and zero-day attack resilience.

Vehicle CAN data contains sensitive information about vehicle status and driving behavior, making it difficult to share directly due to privacy concerns. However, most existing IDS

methods rely on centralized training that requires uploading local data to a central server, posing privacy risks and limiting practical deployment. Federated Learning (FL) offers a promising alternative by enabling local model training with only parameter sharing for global aggregation [5], [6]. This approach balances privacy and collaboration and has been shown to be effective for IDS tasks in prior studies [7].

However, most current FL frameworks still rely on a central server for model coordination and aggregation, which present practical limitations in dynamic and intermittently connected vehicular environments, including communication bottlenecks and a central point of failure issues [8]. In contrast, vehicles naturally support vehicle-to-vehicle (V2V) communication, making it feasible to form decentralized network structures. Notably, CCID-CAN leverages blockchain and cross-chain attack log sharing to realize collaborative in-vehicle intrusion detection in a decentralized manner [9]. This motivates the design of FL architectures that operate without a central server. Recently, Decentralized federated learning (DFL) has attracted increasing attention, where models are exchanged among clients through various decentralized topologies to achieve collaborative training [10]. This paradigm is well aligned with the communication and deployment characteristics of vehicular networks. Although DFL has recently been investigated in vehicular scenarios [11], [12], existing studies are not tailored to time-series CAN intrusion detection and do not sufficiently address performance degradation induced by decentralized model aggregation.

In this work, we aim to address the above limitations. Our main contributions are twofold:

- 1) We propose DCANFL, a decentralized federated learning method for in-vehicle CAN bus intrusion detection. This method is based on the gossip protocol and performs node pairing by calculating model similarity to enable fully decentralized collaborative training. It also integrates a continual learning mechanism to effectively mitigate the performance degradation caused by the decentralized framework.
- 2) We apply the proposed method to a signal-based unsupervised intrusion detection framework and conduct experimental comparisons with the centralized federated learning baseline, FedAvg, as well as other DFL baselines. The results demonstrate that, even without any central server coordination, DCANFL achieves competitive or superior performance compared to the centralized FedAvg baseline, while substantially surpassing existing DFL approaches.

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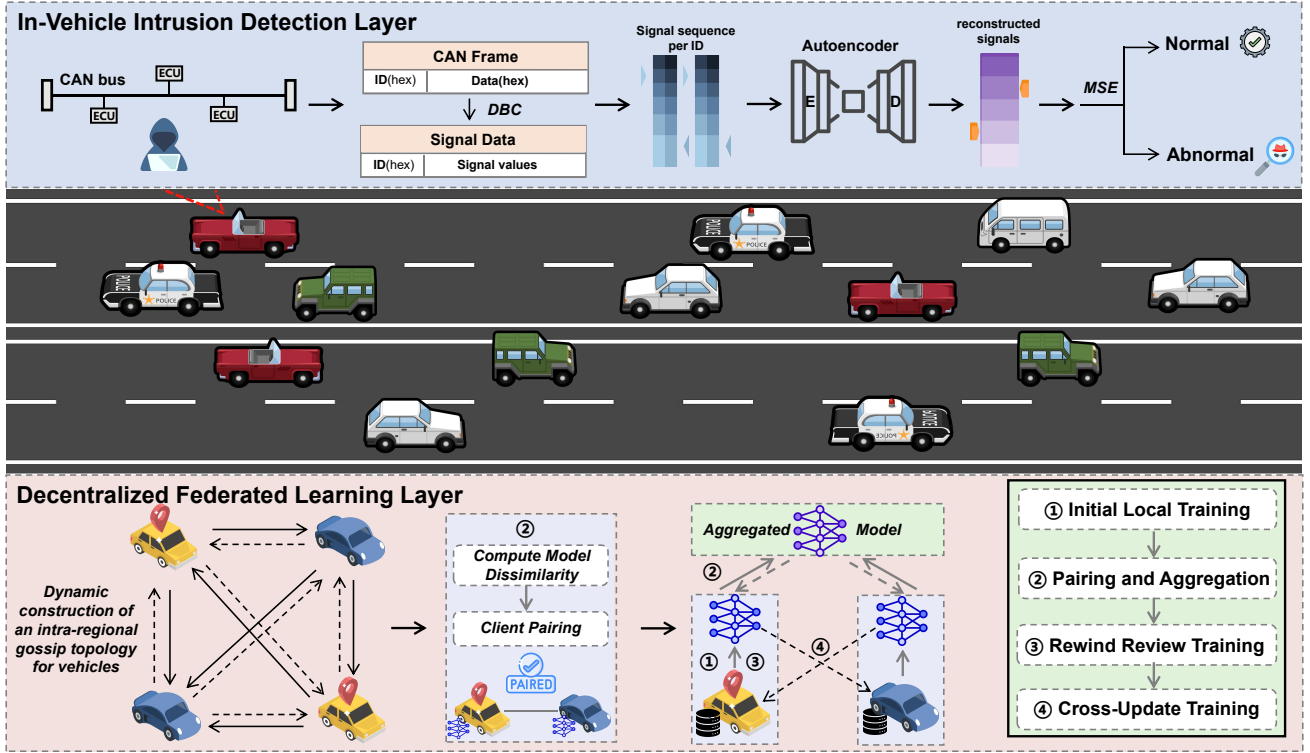


Fig. 1. System model.

II. SYSTEM MODEL

The system model of the proposed decentralized in-vehicle intrusion detection scheme is illustrated in Fig. 1. The scheme consists of a set of intelligent connected vehicles, which operate as decentralized clients in a federated learning paradigm. Each vehicle is equipped with an in-vehicle intrusion detection system and a V2V communication module. Each client $k \in \{1, 2, \dots, K\}$ maintains a local CAN signal dataset $\mathcal{D}_k$ and hosts an intrusion detection model w_k . The objective of the system is to collaboratively train intrusion detection models across clients while preserving the privacy of local data. The overall system architecture is composed of the following two functional layers:

- **In-Vehicle Intrusion Detection Layer:** The in-vehicle intrusion detection layer comprises participating vehicles. Each vehicle performs real-time monitoring of its CAN bus traffic and conducts anomaly detection based on signal reconstruction. Specifically, raw CAN frames are parsed using a DBC file to extract semantically meaningful signal values [13], which are then fed into a locally deployed autoencoder model w_k . The model reconstructs the input signals, and the reconstruction error is used to determine whether the input is anomalous. This layer adopts an unsupervised learning strategy that requires no manual labeling and supports lightweight, real-time intrusion detection.
- **Decentralized Federated Learning Layer:** At this layer, the system employs the Gossip protocol to enable decentralized communication. Spatially neighboring vehicles are grouped into dynamic intra-regional clusters,

and, due to vehicle mobility, the membership of each cluster changes over time. Each vehicle connects to its nearby neighbors to form a lightweight, dynamic gossip graph, over which peer-to-peer (P2P) model training and aggregation are carried out. To improve the efficiency of information exchange, we adopt a dissimilarity-based pairing strategy that prioritizes clients with larger model divergence. To mitigate performance degradation and unstable convergence in conventional DFL, we further design a local training procedure that integrates continual learning, thereby enhancing training stability and overall model performance, as detailed in Section III.

This architecture is built upon a DFL mechanism, enabling vehicles to collaboratively train models through direct V2V communication. By eliminating dependence on a central server, the system achieves greater scalability and adaptability. While preserving the privacy of vehicle data, it supports efficient knowledge sharing and collaborative model construction across nodes. With an unsupervised learning approach, each vehicle is capable of independently performing real-time intrusion detection.

III. THE PROPOSED DCANFL SCHEME

The proposed DCANFL method adopts a signal-based intrusion detection approach at the in-vehicle side and designs a decentralized federated learning scheme to enable collaborative training among vehicles.

A. Signal-based Unsupervised Intrusion Detection

With the increasing diversity of attack strategies in in-vehicle networks, intrusion behaviors often exhibit complex

and unpredictable patterns. Traditional supervised learning methods, which rely heavily on labeled attack data, struggle to generalize to unseen attacks and thus face limitations in practical applications. Furthermore, raw CAN bus data are typically transmitted in hexadecimal format with complex structures and limited semantic meaning, making it challenging to interpret and model effectively.

To address these challenges, the proposed method adopts a signal-based unsupervised learning strategy that enables effective anomaly detection without requiring labeled data. Specifically, the system leverages a DBC file to deserialize the raw payloads in CAN messages, converting the hexadecimal data into meaningful multi-dimensional signal values [13]. These decoded signals reflect the physical state and operational dynamics of various vehicle subsystems, providing discriminative input features for the detection model.

During training, the autoencoder is optimized in an unsupervised manner to reconstruct all signal dimensions under normal conditions. The reconstruction loss is defined as the mean squared error (MSE) between the original and reconstructed signal vectors:

$$\mathcal{L}(\mathbf{x}, \hat{\mathbf{x}}) = \|\mathbf{x} - \hat{\mathbf{x}}\|^2, \quad (1)$$

where $\mathbf{x}$ and $\hat{\mathbf{x}}$ denote the original and reconstructed signal vectors, respectively, and $\|\cdot\|$ denotes the ℓ_2 norm. This loss computes the error across all signals at each time step, thereby capturing inter-signal dependencies. In the detection phase, the same reconstruction error $\mathcal{L}$ is used as the anomaly score, and samples whose reconstruction errors exceed a predefined threshold are flagged as anomalies.

B. Gossip Protocol with Dissimilarity-driven Pairing

In the proposed DCANFL framework, we adopt the Gossip protocol as a decentralized communication mechanism. Unlike centralized federated learning that requires all clients to upload models to a central server, Gossip enables each vehicle node to perform P2P communication and aggregation. This decentralized design improves scalability, fault tolerance, and resilience to network failures.

To further enhance information diversity and accelerate convergence, we replace purely random pairing with a dissimilarity-driven strategy. In each communication round, neighboring clients first compute the pairwise cosine similarities between their current local models. The cosine similarity between clients i and j at round t is defined as

$$\text{sim}_{i,j}^t = \frac{\langle \mathbf{w}_i^t, \mathbf{w}_j^t \rangle}{\|\mathbf{w}_i^t\| \cdot \|\mathbf{w}_j^t\|}, \quad (2)$$

where $\mathbf{w}_i^t$ and $\mathbf{w}_j^t$ denote the local models of clients i and j at round t , $\langle \cdot, \cdot \rangle$ represents the inner product, and $\|\cdot\|$ denotes the Euclidean ℓ_2 norm.

To construct the aggregation pairs, we adopt a dissimilarity-driven greedy strategy. Specifically, all possible client pairs (i, j) with $i < j$ and $i, j \in \mathcal{K}$ are first sorted in ascending order based on their cosine similarity $\text{sim}_{i,j}^t$. Then, the pairing set $\mathcal{S}$ is constructed by iteratively selecting the most dissimilar pairs, ensuring that each client appears in at most one pair. If

the number of clients is odd, the remaining unpaired client performs only local training in that communication round. Formally, the pair selection process can be expressed as

$$\mathcal{S} = \{(i, j) \in \mathcal{C} \mid \text{sim}_{i,j}^t \text{ is minimal and } i, j \notin \mathcal{S}\}, \quad (3)$$

where $\mathcal{C} = \{(i, j) \mid i < j, i, j \in \mathcal{K}\}$ denotes the set of all candidate pairs.

C. Federated Continual Learning Based on Gossip

Compared with centralized federated learning, gossip-based federated learning is more susceptible to two forms of heterogeneity. On the systemic side, time-varying V2V connectivity and neighborhood relationships, together with partial client participation in each round, lead to imbalanced information propagation and inconsistent optimization trajectories, which further exacerbate unstable training dynamics and increase convergence variance. On the statistical side, the non-stationarity of temporal CAN messages induces time-varying data distributions and cross-client distribution shift, resulting in pronounced statistical heterogeneity. The interplay of these two heterogeneity sources often causes performance degradation and yields larger convergence variance. To mitigate these issues, we introduce continual learning [14] and design a local training procedure embedded into the gossip communication mechanism described in Section III-B, so that all gradients are computed locally while only model parameters are exchanged. This lightweight rehearsal over previously encountered partner distributions mitigates cross-client distribution shift and counteracts the representation drift caused by weighted parameter averaging. Overall, for CAN messages with pronounced temporal characteristics, this rehearsal-style local training mechanism is simple yet effective in improving training stability and preserving key temporal patterns. In each communication round, the training process consists of the following four stages:

1) Initial Local Training: Each client k updates its local model $\mathbf{w}_k^t$ on the local dataset $\mathcal{D}_k$ for e epochs, as follows

$$\mathbf{w}_k^t = \mathbf{w}_k^{t-1} - \eta \sum_{s=0}^{e-1} \nabla \mathcal{L}(\mathbf{w}_k^{t-1}, \mathcal{D}_k), \quad (4)$$

where $\mathbf{w}_k^t$ denotes the updated model of client k at round t , η is the learning rate, $\nabla \mathcal{L}$ denotes the gradient of the local loss function, and $\mathcal{D}_k$ is the local dataset of client k .

2) Gossip Pairing and Weighted Aggregation: Clients are paired as (i, j) based on model dissimilarity, as described in Section III-B. The weighted aggregation is then performed as

$$\mathbf{w}_{ij}^t = \frac{n_i \mathbf{w}_i^t + n_j \mathbf{w}_j^t}{n_i + n_j}, \quad (5)$$

where $\mathbf{w}_i^t$ and $\mathbf{w}_j^t$ denote the local models of clients i and j at round t , respectively, and n_i, n_j represent the number of local training samples at clients i and j .

3) Rewind Review Training: For each paired clients (i, j) , the aggregated model $\mathbf{w}_{ij}^t$ is computed locally at both clients, and further trained for $E - 2e$ epochs on their own datasets. For example, on client j , $\mathbf{w}_{ij}^t$ is updated on $\mathcal{D}_j$ to obtain $\tilde{\mathbf{w}}_j^t$,

which serves as the rewind-review model prepared for client i and will be returned to client i in the next step.

$$\tilde{\mathbf{w}}_i^t = \mathbf{w}_{ij}^t - \eta \sum_{s=0}^{E-2e-1} \nabla \mathcal{L}(\mathbf{w}_{ij}^s, \mathcal{D}_j), \quad (6)$$

$$\tilde{\mathbf{w}}_j^t = \mathbf{w}_{ij}^t - \eta \sum_{s=0}^{E-2e-1} \nabla \mathcal{L}(\mathbf{w}_{ij}^s, \mathcal{D}_i). \quad (7)$$

4) Cross-Update Training: After the rewind-review step, the matched client pair exchange the reviewed models $\tilde{\mathbf{w}}_i^t$ and $\tilde{\mathbf{w}}_j^t$ produced by their partners. Each client then trains the received model on its own dataset for e epochs, thereby completing this communication round, as given by

$$\mathbf{w}_i^t = \tilde{\mathbf{w}}_i^t - \eta \sum_{s=0}^{e-1} \nabla \mathcal{L}(\tilde{\mathbf{w}}_i^s, \mathcal{D}_i), \quad (8)$$

$$\mathbf{w}_j^t = \tilde{\mathbf{w}}_j^t - \eta \sum_{s=0}^{e-1} \nabla \mathcal{L}(\tilde{\mathbf{w}}_j^s, \mathcal{D}_j). \quad (9)$$

This mechanism mitigates decentralization-induced performance degradation through a redesigned training procedure summarized in Algorithm 1, where $\text{Train}(\mathbf{w}, \mathcal{D}, e)$ denotes the local routine that runs e epochs of optimization on dataset $\mathcal{D}$ starting from parameters $\mathbf{w}$.

Algorithm 1 DCANFL: Decentralized Federated Learning for In-Vehicle CAN Intrusion Detection

- 1: **Input:** Local datasets $\{\mathcal{D}_k\}$ for all clients $k = 1, \dots, K$, epochs per stage e , total epochs per round E , total rounds T
 - 2: **Output:** Final decentralized models $\{w_k^T\}$ for all clients
 - 3: **for** each round $t = 1$ to T **do**
 - 4: Select available clients $\mathcal{A}_t$ at round t ;
 - 5: **for** each client $k \in \mathcal{A}_t$ **in parallel do**
 - 6: Initial Local Training: $\mathbf{w}_k \leftarrow \text{Train}(\mathbf{w}_k, \mathcal{D}_k, e)$;
 - 7: **end for**
 - 8: Compute all pairwise similarities $\text{sim}_{i,j}^t$;
 - 9: Greedy form pairing set $\mathcal{S}$ by sorting $\text{sim}_{i,j}^t$ in ascending order;
 - 10: **for** each pair $(i, j) \in \mathcal{S}$ **in parallel do**
 - 11: Compute sample sizes: $n_i = |\mathcal{D}_i|$, $n_j = |\mathcal{D}_j|$;
 - 12: Aggregate models using (5) to get $\mathbf{w}_{ij}^t$;
 - 13: Rewind Review Training using (6) and (7):
 - 14: $\tilde{\mathbf{w}}_i^t \leftarrow \text{Train}(\mathbf{w}_{ij}^t, \mathcal{D}_j, E - 2e)$;
 - 15: $\tilde{\mathbf{w}}_j^t \leftarrow \text{Train}(\mathbf{w}_{ij}^t, \mathcal{D}_i, E - 2e)$;
 - 16: Exchange updated models between paired clients;
 - 17: Cross-Update Training using (8) and (9):
 - 18: $\mathbf{w}_i^t \leftarrow \text{Train}(\tilde{\mathbf{w}}_i^t, \mathcal{D}_i, e)$;
 - 19: $\mathbf{w}_j^t \leftarrow \text{Train}(\tilde{\mathbf{w}}_j^t, \mathcal{D}_j, e)$;
 - 20: **end for**
 - 21: **end for**
 - 22: **return** $\{\mathbf{w}_k^T\}$;
-

To characterize the communication overhead and scalability of DCANFL, we summarize its per-round communication cost as follows. Let P denote the number of model parameters,

K the number of active clients participating in each round within a cluster, and d the number of peers with whom a client exchanges model information during the similarity-estimation stage. Let R denote the size of the exchanged model representation used for similarity estimation, where $R \leq P$ and $R = P$ corresponds to exchanging full model parameters. In centralized FedAvg, each active client uploads one local model and downloads one aggregated global model per round, incurring about $2P$ model-sized transmissions and thus $\mathcal{O}(P)$ communication complexity. In DCANFL, similarity-based pairing requires exchanging the representation of size R with d peers to compute cosine similarity, which incurs about $2dR$ model-sized transmissions per client per round in the worst case. After pairing, each client exchanges the rewind-review model with its matched peer for cross-update training, incurring an additional $2P$ model-sized transmissions per round. Therefore, the per-round per-client communication cost of DCANFL is bounded by $2dR + 4P$. Since DCANFL operates within clusters, d is typically bounded by the local neighborhood size, which keeps the practical overhead manageable. Moreover, DCANFL enables fully decentralized training without a central server and distributes traffic over local V2V links, avoiding server-side bottlenecks and improving deployability and resilience in vehicular networks.

IV. EXPERIMENT

A. Experiment Setup

We used two representative CAN datasets, SynCAN [2] and X-CANIDS [3], to evaluate the proposed intrusion detection methods. SynCAN is a widely used benchmark with synthetic attacks, while X-CANIDS contains real CAN traffic from a production vehicle, which we deserialized into signal-level representations using OpenDBC [13]. Both datasets cover five attack types for comprehensive evaluation. X-CANIDS further assesses typical IDs (e.g., 2B0h, 044h, 47fh, 220h) under suspension, fabrication, and masquerade attacks, and includes fuzzing attacks of varying intensities (fuzz-10 to fuzz-1000). Models were trained unsupervised using the revised CANet architecture [4], capturing temporal patterns without labeled data. A validation split was used to set detection thresholds, which were then applied to the test set for performance evaluation across attack scenarios.

The experiments were implemented using the PyTorch framework. All models were trained using the mean squared error loss with a fixed learning rate of 1×10^{-4} and a batch size of 256. In the experiments, the number of FL clients was set to 10. These 10 clients belong to a single cluster, and the CAN frames are first sorted in chronological order and sliced into 10 non-overlapping segments to construct their local datasets, while all FL methods share the same client data split and hyperparameters. The number of local training epochs per stage e was set to 1, and the total number of epochs per round E was set to 3. The number of available clients participating in each round ranged from 7 to 10. The total number of communication rounds T was set to 15 for the SynCAN dataset and 20 for the X-CANIDS dataset.

In the experiments, we evaluated the performance of FedAvg [5], DCANFL, DFL [10], and GL [15] on the intrusion

TABLE I
COMPARISON OF DIFFERENT FL METHODS ACROSS 10 CLIENTS ON THE SYNCAN DATASET

FL Method	Flooding		Suppress		Plateau		Continuous		Playback		Average	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
FedAvg	0.9859	0.9457	0.8656	0.7311	0.9858	0.9106	0.9435	0.8352	0.9803	0.9305	0.9526	0.8706
GL	0.9368	0.7692	0.8229	0.5621	0.9366	0.7317	0.9213	0.7219	0.9663	0.7875	0.9168	0.7145
DFL	0.9255	0.8487	0.8319	0.6333	0.9238	0.8320	0.9301	0.7814	0.9720	0.9052	0.9166	0.8001
DCANFL	0.9829	0.9324	0.8692	0.7393	0.9866	0.9112	0.9562	0.8691	0.9814	0.9354	0.9553	0.8775

TABLE II
COMPARISON OF DIFFERENT FL METHODS ACROSS 10 CLIENTS ON THE X-CANIDS DATASET

FL Method	Fuzzing		Fabrication		Masquerade		Suspension		Replay		Average	
	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1
FedAvg	0.9000	0.7938	0.8103	0.7209	0.8470	0.7361	0.5715	0.4202	0.8742	0.8981	0.8006	0.7138
GL	0.8462	0.6854	0.7374	0.6647	0.8582	0.7328	0.5288	0.5389	0.8416	0.7933	0.7624	0.6830
DFL	0.8864	0.7218	0.7886	0.6673	0.8104	0.6889	0.6666	0.5363	0.8689	0.8361	0.8042	0.6901
DCANFL	0.8980	0.7870	0.8107	0.7409	0.8582	0.7623	0.5681	0.5290	0.9055	0.8846	0.8081	0.7408

detection task across 10 clients. The evaluation metrics were the AUC and F1 scores. AUC measures the trade-off between true and false positive rates, while F1 is the harmonic mean of precision and recall. Both are standard in intrusion and anomaly detection and capture overall detection performance and the balance between missed and false alarms.

B. Experiment Results

Table I presents the detection performance of different FL methods on the SynCAN dataset, covering five typical attack types. In all scenarios, DCANFL consistently achieves high AUC and F1 scores. On average, DCANFL attains an overall AUC of 0.9553 and F1 score of 0.8775, slightly outperforming FedAvg and significantly surpassing the decentralized FL methods GL and DFL, demonstrating consistently high detection performance across different attack types and clients, as well as strong adaptability in simulated vehicular anomaly detection tasks.

Table II reports the evaluation results on the real-world X-CANIDS dataset. Compared to the simulated environment, the real-world dataset involves more noise, dynamic behaviors, and signal diversity. Despite these challenges, DCANFL maintains strong performance, achieving the highest F1 scores across several key attack types. Its overall average F1 score reaches 0.7408, outperforming all baseline methods, further validating the generalization capability and effectiveness of the proposed method in complex real-world conditions.

It is worth emphasizing that in the baseline FedAvg setting, a central server is assumed to perform global aggregation, whereas DCANFL operates in a fully decentralized manner without relying on any central server coordination during training. While retaining the advantages of decentralized communication, it significantly outperforms the GL method, which only adopts random gossip. The comparison with GL can also be regarded as an ablation study. These results strongly validate the effectiveness and practical value of our design, particularly the model dissimilarity-driven pairing strategy

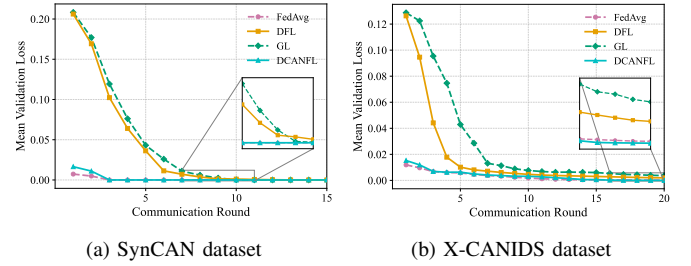
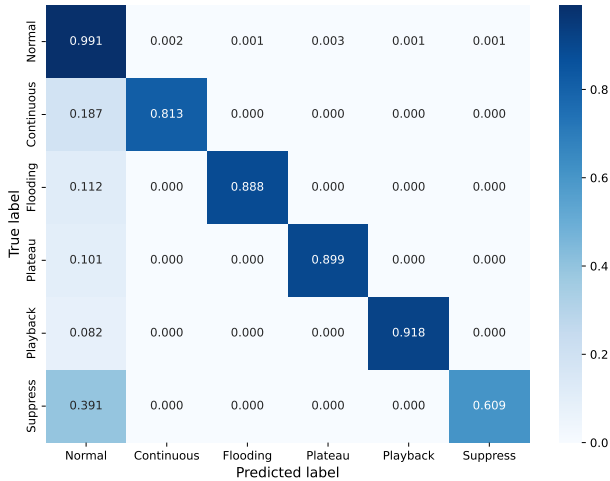


Fig. 2. Convergence comparison of different FL methods on two intrusion detection datasets. The validation MSE is evaluated at each FL communication round.

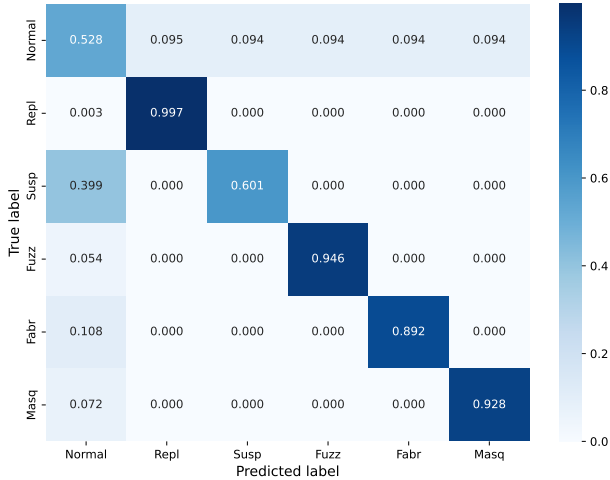
and the continual learning mechanism, which play a critical role in enhancing collaborative training efficiency and model performance.

Fig. 2 illustrates the convergence behavior of different FL methods on the SynCAN and X-CANIDS intrusion detection datasets. For a fair comparison, the validation MSE is computed at the end of each FL communication round on the corresponding validation set of each dataset. It can be observed that the proposed DCANFL framework and FedAvg both converge substantially faster on the two datasets, and clearly outperform DFL and the GL method with random gossip. In intrusion detection tasks, the MSE directly quantifies the reconstruction or prediction deviation between the model output and normal traffic patterns, and therefore serves as a sensitive indicator of anomaly discrimination capability. The faster decrease in validation MSE achieved by DCANFL thus provides quantitative evidence of its superior convergence efficiency in DFL scenarios.

Fig. 3 presents the confusion matrix visualizations of our proposed DCANFL method on both datasets. In both cases, DCANFL exhibits clear diagonal dominance across most attack types, indicating strong intrusion detection capabilities. A notable exception is the Suspension attack, which shows significant confusion with the Normal class. This is attributed



(a) Confusion matrix on SynCAN dataset



(b) Confusion matrix on X-CANIDS dataset

Fig. 3. Confusion matrices of DCANFL on the SynCAN and X-CANIDS datasets.

to the difficulty of the CANet model in detecting Suspension attacks due to its high sensitivity to signal variations. Since such attacks often involve minimal or no signal change, distinguishing between naturally stable signals and halted updates remains challenging, consistent with prior findings [4]. Nevertheless, for other attack types with more distinct signal patterns, DCANFL continues to achieve high detection performance under the 10-client FL setting.

V. CONCLUSION

This paper proposed DCANFL, a decentralized federated learning method for in-vehicle CAN intrusion detection. It employs gossip-based model similarity pairing for V2V collaboration without a central server and integrates continual learning to mitigate performance degradation. Experiments show that DCANFL outperforms FedAvg and other baselines under fully decentralized settings. Nevertheless, further validation and refinement are needed in multi-vehicle-type and highly dynamic V2V scenarios, and more comprehensive security safeguards are required to protect model updates

and gradient-level information. Future work will investigate model heterogeneity across different vehicle types, resilience to Byzantine nodes and highly dynamic V2V mobility, and the integration of techniques such as differential privacy to further enhance privacy and security for in-vehicle applications.

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